

Innocent Kisoka

Date of birth: 18/03/1999 | **Nationality:** Tanzanian | **Phone number:** (+41) 0772279134 (Mobile) | **Email address:** kisoki@usi.ch | **Website:** https://innocentkisoka.github.io |

Address: Via Zurigo 1 Lugano , 6900, Lugano, Switzerland (Home)

ABOUT ME

- MSc Artificial Intelligence student at USI with focused research experience in Large Language Models, preference optimisation, controllability of latent representations, and generative AI.
- Currently extending the CRPO framework to low-resource languages to improve alignment, generalisation, and creative capabilities of LLMs.
- Strong technical foundation in attention mechanisms, representation learning, and scalable experimentation. Eager to contribute to LLM and agentic model research.

EDUCATION AND TRAINING

16/09/2024 – CURRENT Lugano, Switzerland

MASTER'S IN ARTIFICIAL INTELLIGENCE Università della Svizzera italiana (USI)

- **Master's Thesis (ongoing)**: Extending Creative Preference Optimization (CRPO) to Swahili – preference optimisation of latent representations for controllability and generalisation in low-resource models
- **Relevant Coursework**: Deep Learning Lab, Graph Deep Learning, Machine Learning, Scientific Machine Learning, High-Performance Computing, Numerical Algorithms, Natural Language Processing
- **Focus**: Representation learning, Graph Neural Networks, Scientific Machine Learning, Parallel Computing

Website https://www.usi.ch/en

01/09/2020 – 20/08/2024 Sokcho, South Korea

BACHELOR OF COMPUTER ENGINEERING Kyungdong University

Website https://global.kduniv.ac.kr/global/index.php |

Thesis Enhancing Email Classification through Machine Learning Algorithms

01/07/2017 – 31/05/2019 Morogoro, Tanzania

SECONDARY SCHOOL Mzumbe Secondary School, Morogoro, Tanzania.

Address P.O.Box 19, 67311, Morogoro, Tanzania

SKILLS

Python (NumPy, Pandas, Matplotlib) | Microsoft Office | Sql | C# | C++ | Html | Java | Parallel programming frameworks (MPI, OpenMP, OpenCL) - Basics | Strong knowledge of parallel programming (MPI and threads) | Generative AI, LLMs, Deep Learning, NLP, Statistical Modeling | Financial AI Applications: Algorithm Development, AI Ethics & Fairness, Data Analysis | PyTorch Geometric | PyTorch

LANGUAGE SKILLS

Mother tongue(s): **SWAHILI**

Other language(s): **ITALIAN**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1
KOREAN	B2	B2	B2	B2	B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

ACHIEVEMENTS

Recognitions and Grants

2024: The Foundation for the Lugano Faculties of the Università della Svizzera italiana - Switzerland
2022: Programming Excellence Award (Kyungdong University)
2020: Kdu Global - IUCEA -100% Scholarship (2nd Cohort) - South Korea
2017: Academic Excellence Award (Mzumbe Secondary School)
2016: Academic Excellence Award (Rosmini Secondary School)

HOBBIES

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Football, Gym, Hiking, Travelling

QUALITIES

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Team worker, Diligent, Compassionate

PROJECTS

Projects & Research

Graph Neural Networks Use Graphs When They Shouldn't (ICML 2024 Replication Study)
- Led rigorous replication of the R-COV method for mitigating topological overfitting in attention-based Message-Passing GNNs.
- Investigated how graph structure influences latent representations and model behaviour even when topology is non-informative; analysed implicit bias through weight-ratio dynamics and PCA visualisation of clusters.
- Demonstrated the critical role of graph regularity on representation quality- implemented custom attention mechanisms and edge-feature augmentation in PyTorch Geometric.
- Results directly relevant to interpretability and controllability of latent representations.

Traveling Salesman Problem with Transformers
- Developed Transformer-based heuristic using self-attention for combinatorial optimisation.

Text Generation with LSTMs
- Designed LSTM-based generative models for coherent political news titles; evaluated fluency, coherence, and controllability of latent representations.

Cardiac Wave Propagation PDE Solver – Hybrid FEM + PINN
- Designed and implemented a mesh-free Physics-Informed Neural Network for solving anisotropic diffusion-reaction PDEs, learning latent representations of physical fields under varying diffusion coefficients.
- Benchmarked against classical IMEX Finite Element Method; demonstrated generalisation and interpretability of learned representations.

Parallel Solutions for Nonlinear PDEs
- Implemented **parallel PDE solvers** using **MPI and OpenMP**
- Optimized computational efficiency for large-scale simulations

Link https://github.com/InnocentKisoka

RESEARCH INTERESTS

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- Large Language Models and low-resource NLP
- Agentic models and multi-step reasoning in LLMs
- Preference optimisation and alignment for controllable generation
- Representation learning and interpretability in deep learning systems
- Vision-Language Models and multimodal intelligence

REFEREES

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Prof. Fabio Crestani
Università della Svizzera italiana (USI)

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East Campus, Sector D, Office D2.13 (Level P2) Via la Santa 1, 6962 Lugano-Viganello, Switzerland

Prof. Stefan Wolf
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East Campus, Sector D, Office D5.06 (Level P5) Via la Santa 1, 6962 Lugano-Viganello, Switzerland

Prof. Barry Smith, PhD, FACMI, FIAOA Professor of Philosophy, Biomedical Informatics, and Computer Science and Engineering
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